# MILITARY STANDARD

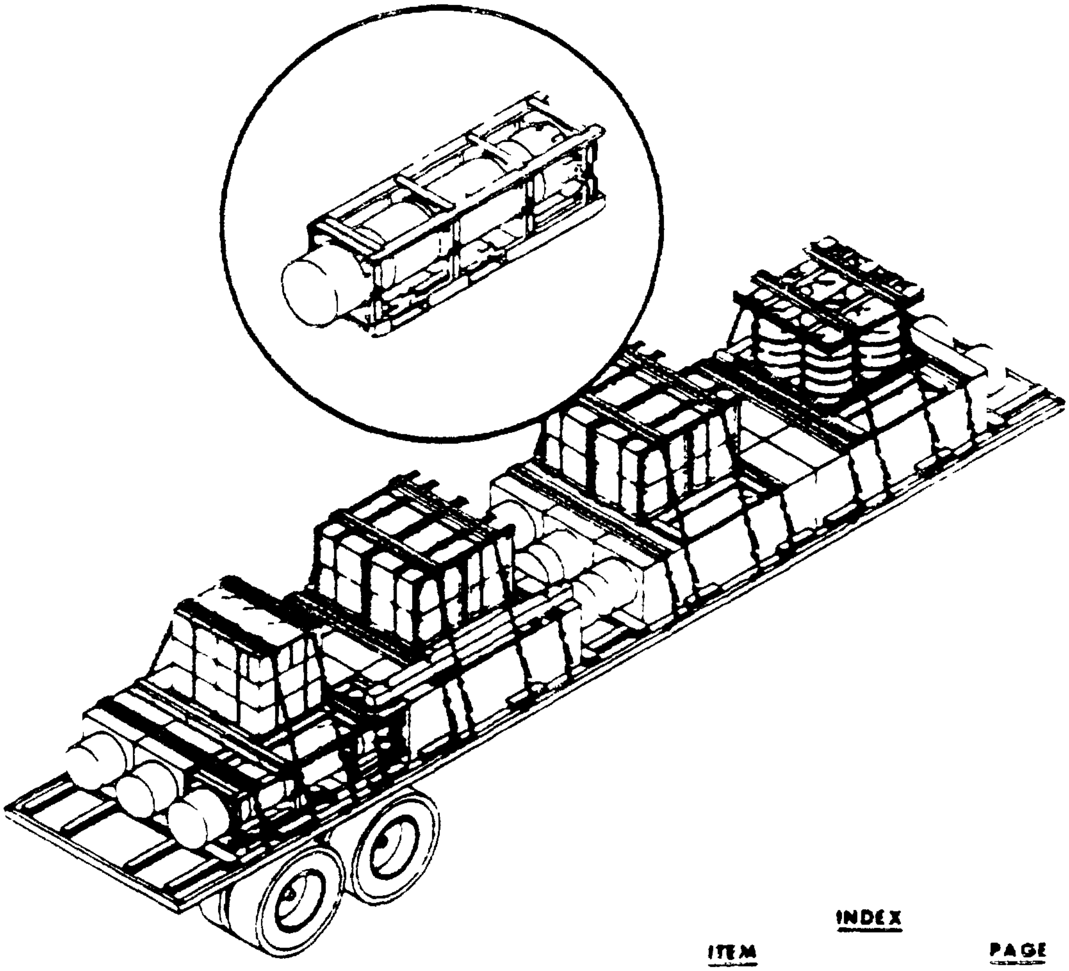
MIL-STD-1320-166
(NAVY)

30 SEPTEMBER 1975

## TRUCKLOADING OF HAZARDOUS MATERIALS
## MINE, UNDERWATER MK 56 MOD 0 IN CRATE MK 56
## "CONFIGURATION B" SUB ASSEMBLY
## WITH ASSOCIATED FLIGHT GEAR

**UNIT LOAD DATA**

SEE TABLE I,
PAGE 2



NOTES

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINER-LOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-1320 (NAVY)

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FSC 8140

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M.S. Gray NWHL 9/15/75
SIGNATURE [illegible] DATE

J.E. Kelly 9/16/75
SIGNATURE [illegible] DATE

**ORIGINATOR** 9/15/75
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

CODE I D [illegible]

0231

F-33

# GENERAL NOTES

THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE MK 56 MOD 0 UNDERWATER MINE "CONFIGURATION B" IN CRATE MK 56 MOD 0, FOUR DOMESTIC UNIT LOADS OF RELATED FLIGHT GEAR AND ONE WOOD BOX OF ARMING WIRES (SEE TABLE I BELOW).

**NOTE**

THE FLIGHT GEAR SHALL BE PALLETIZING IN ACCORDANCE WITH THE PALLETIZING WR'S LISTED IN TABLE I. THE UNIT LOADS OF FLIGHT GEAR MAY BE OMITTED FROM THE TRUCKLOAD OR ANY COMBINATION OF FLIGHT GEAR UNIT LOADS MAY BE LOADED. THE CORRECT PALLET BASE (DETAILS C OR D) SHALL BE USED FOR THE UNIT LOAD BEING SHIPPED.

THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 FT FLATBED TRAILERS WITH THE TRAILERS' TANDEM AXLES LOCATED IN THE "WESTERN POSITION" (AT THE EXTREME REAR OF THE TRAILER). DO NOT USE TRAILERS WITH METAL FLOORS.

3 THE MINES ONLY SHALL BE COVERED WITH A FIRE RESISTANT, WATERPROOF TARPAULIN. MINES SHALL BE COVERED WITH THE TARPAULIN PRIOR TO APPLYING THE TIE-DOWN CHAINS. PROTECTOR BOARDS SHALL BE USED BETWEEN THE TARPAULIN AND THE CHAINS. TARPAULINS ARE NOT REQUIRED FOR THE FLIGHT GEAR. THE LOAD SHOWN ON PAGE 3 HAS HAD THE TARPAULIN OMITTED FOR CLARITY.

4 THE UNIT LOADS IN THE SECOND LAYER SHALL BE POSITIONED ON THE APPROPRIATE PALLET BASE, SEE DETAILS C AND D. THE PALLET BASE IS POSITIONED ON TOP OF THE TARPAULIN COVERED MINES AND LOCATED SO THAT IT LOCKS OVER THE MINES CRATE TOP CENTER TWO CROSSMEMBERS.

5 CHAINS AND LOAD BINDERS SHALL BE USED FOR TIEDOWNS TO SECURE THE LOAD TO THE TRAILER AND THEY SHALL MEET THE FOLLOWING REQUIREMENTS:

a. CHAINS AND FITTINGS SHALL BE 3/8 INCH AND LOADBINDERS 3/8 INCH CAPACITY.

b. CHAINS, FITTINGS AND LOADBINDERS SHALL HAVE A MINIMUM SAFE WORKING LOAD OF 5000 LBS.

"LEBUS" TYPE LOADBINDERS SHALL BE SAFETY WIRED WITH 16 GAUGE SOFT ANNEALED IRON WIRE OR SECURED USING THE END PIECE OF THE 3/8 CHAIN.

6 PRIOR TO LOADING THE TRAILER AND DURING PRELOADING INSPECTION REQUIRED BY NAVWEPS OP 2165 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS AND LOADBINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS AND LOADBINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE ON ITEM NO. 22 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR A REJECTION OF A CHAIN OR LOADBINDER.

THE LADING SHALL BE PROTECTED FROM CHAIN DAMAGE WHERE THE CHAIN GOES OVER THE MINE CRATE AND THE UNIT LOADS BY INSERTING A 2 X 6 BETWEEN THE CHAIN AND THE UNIT LOAD.

8 THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR TRAILER WILL NOT EXCEED THESE LIMITATIONS.

9 AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO DUNNAGE IN AN ACCESSIBLE AREA AND ATTACH "EXPLOSIVES A" PLACARD TO BOTH SIDES, FRONT AND BACK OF TRAILER.

10 APPLICABLE MATERIAL SPECIFICATIONS:

DUNNAGE LUMBER: MMM-L-751

NAILS: FF-N-105, TYPE II, STYLE 10, COMMON BRIGHT

STRAPPING: QQ-S-781, TYPE I, CLASS A

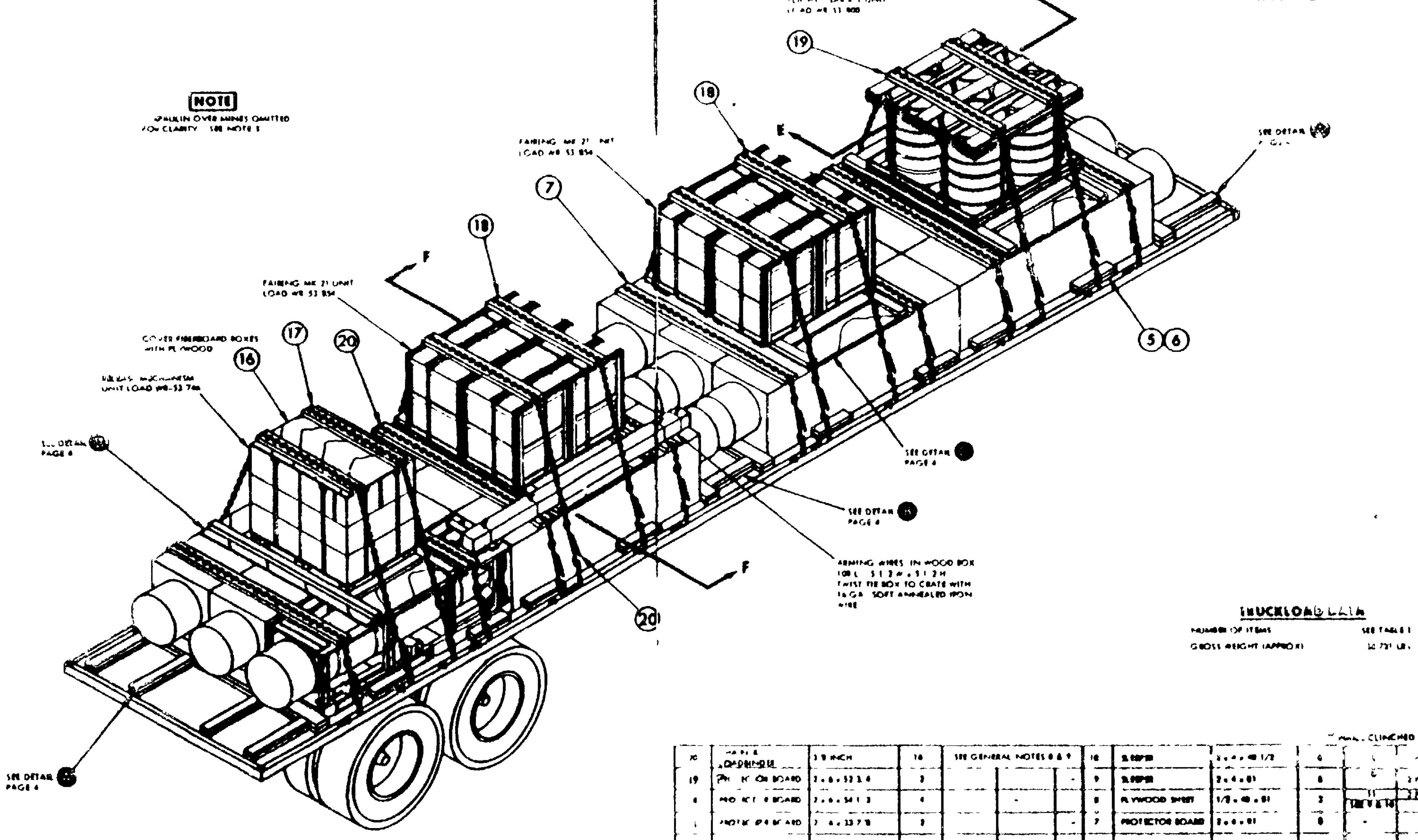
PLYWOOD: COMMERCIAL STANDARD CS45, GRADE EXTERIOR C-C

FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY).

## TABLE I

"CONFIGURATION B" MINE, UNDERWATER, MK 56

| UNIT LOAD TITLE & DOCUMENT NUMBER | NO. OF UNIT LOADS | WEIGHT OF ONE UNIT LOAD (APPROX) | DIMENSIONS OF UNIT LOADS | CUBE OF UNIT LOAD | DOT HAZARD CLASSIFICATION |
|---|---|---|---|---|---|
| UNDERWATER MINE, MK 56 - LDS 30108 | 12 | 2219 LBS | 106 L x 27 W x 29 1/2 H | 48.8 CU FT | EXPLOSIVES A |
| FAIRING, MK 71 - WR-53 836 | 2 | 724 LBS | 54 1/2 L x 54 1/2 W x 48 3/4 H | 78.0 CU FT | INERT |
| FLIGHT GEAR KIT - WR-53 880 | 1 | 1266 LBS | 52 3/4 L x 52 3/4 W x 48 3/8 H | 66.4 CU FT | EXPLOSIVES C |
| RELEASE MECHANISM - WR-53 746 | 1 | 1340 LBS | 47 L x 35 7/8 W x 42 1/4 W | 38.9 CU FT | INERT |
| ARMING WIRES, BOX, WOOD | 1 | 39 LBS | 100 L x 5 1/2 W x 5 1/2 H | 1.9 CU FT | INERT |



**NOTE**
TARPAULIN OVER MINES OMITTED
FOR CLARITY SEE NOTE 3

**TRUCKLOAD DATA**

NUMBER OF ITEMS SEE TABLE I
GROSS WEIGHT (APPROX) 36 731 LB.

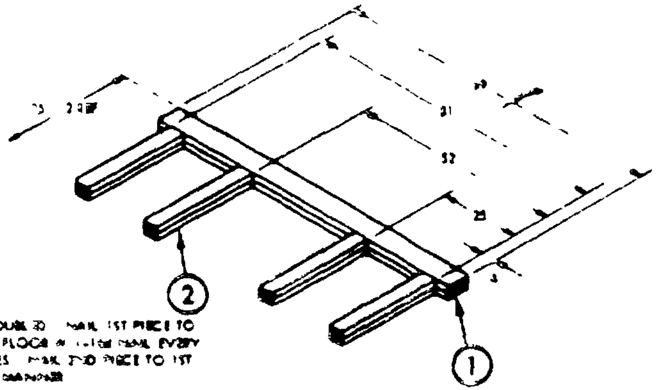
| PIECE NO | DESCRIPTION | SIZE | NO PCS REQD | NAIL TO | NAILS NUMBER | NAILS SIZE |
|---|---|---|---|---|---|---|
| 20 | [illegible] DUNNAGE | 3/8 INCH | 16 | SEE GENERAL NOTES 8 & 9 | | |
| 19 | PROTECTION BOARD | 2 x 6 x 52 3/4 | 2 | | | - |
| 18 | PROTECTOR BOARD | 2 x 6 x 54 1/2 | 4 | | - | |
| 17 | PROTECTOR BOARD | 2 x 6 x 33 7/8 | 2 | | | - |
| 16 | PLYWOOD PANEL | 1/2 x 33 7/8 x 47 | 1 | - | | - |
| 15 | RUNNER | 2 x 6 x 81 | 2 | SEE 13 | | |
| 14 | SLEEPER | 2 x 6 x 36 | 2 | 12 | 2 PER FOOT | 6d |
| 13 | SLEEPER | 2 x 4 x 81 | 2 | 12 | 2 PER FOOT | 6d |
| 12 | PLYWOOD SHEET | 1/2 x 48 x 81 | 1 | 11<br>SEE DETAIL A | 2 PER FOOT<br>- | 6d<br>- |
| 11 | RUNNER | 2 x 6 x 81 | 2 | SEE 8 | | |

LIST OF MATERIALS AND NAILING DATA

NAILS CLINCHED

| PIECE NO | DESCRIPTION | SIZE | NO PCS REQD | NAIL TO | NAILS NUMBER |
|---|---|---|---|---|---|
| 10 | SLEEPER | 2 x 4 x 48 1/2 | 6 | [illegible] | [illegible] FOOT |
| 9 | SLEEPER | 2 x 4 x 81 | 6 | 6 | 2 PER FOOT |
| 8 | PLYWOOD SHEET | 1/2 x 48 x 81 | 2 | 11<br>SEE 9 & 10 | 2 PER FOOT<br>- |
| 7 | PROTECTOR BOARD | 2 x 6 x 81 | 8 | - | |
| 6 | SLEEPER | 2 x 4 x 10 | 16 | 3 | 6 |
| 5 | SLEEPER | 2 x 4 x 10 | 16 | TOENAIL IN PLACE | 6 |
| 4 | STRUT | 2 x 4 x 22 | 16 | SEE DETAIL B, PAGE 4 | |
| 3 | CROSSMEMBER | 2 x 6 x 89 | 4 | | |
| 2 | BACKUP CLEAT | 2 x 4 x 20 | 16 | SEE DETAIL A, PAGE 4 | |
| 1 | CROSSMEMBER | 2 x 6 x 89 | 4 | | |

LIST OF MATERIALS AND NAILING DATA

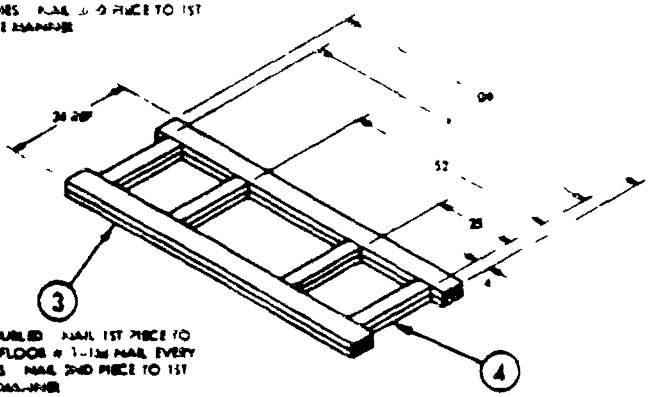


2 x 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W 1-12d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER

2 x 6 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-12d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER

**DETAIL A**

FRONT AND REAR BLOCKING
ASSEMBLE IN PLACE

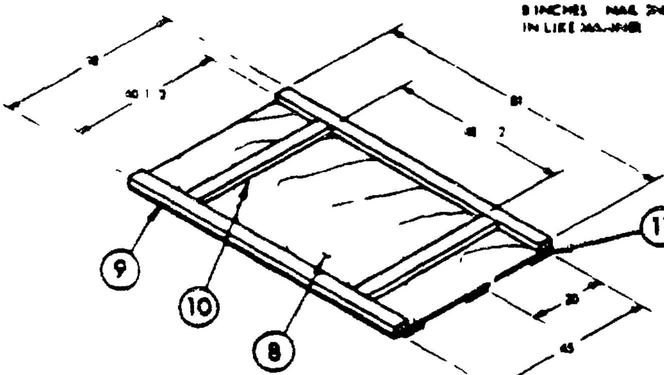


2 x 6 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W 1-12d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER

2 x 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W 6-12d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER

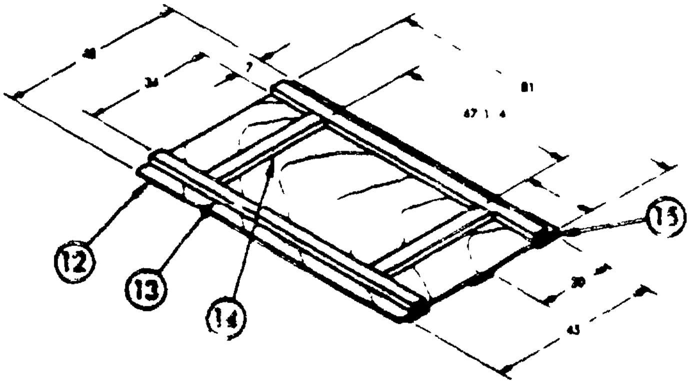
**DETAIL B**

CENTER BLOCKING
ASSEMBLE IN PLACE



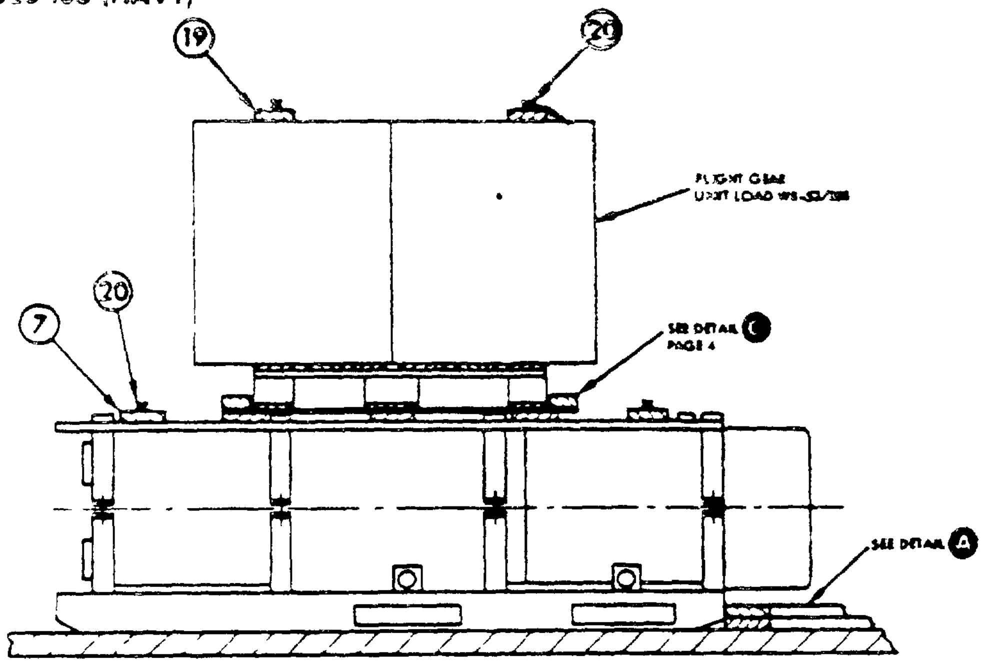
**DETAIL C**

PALLET BASE
(FOR ALL FLIGHT GEAR EXCEPT
PARACHUTE RELEASE MECHANISM)
3 REQUIRED

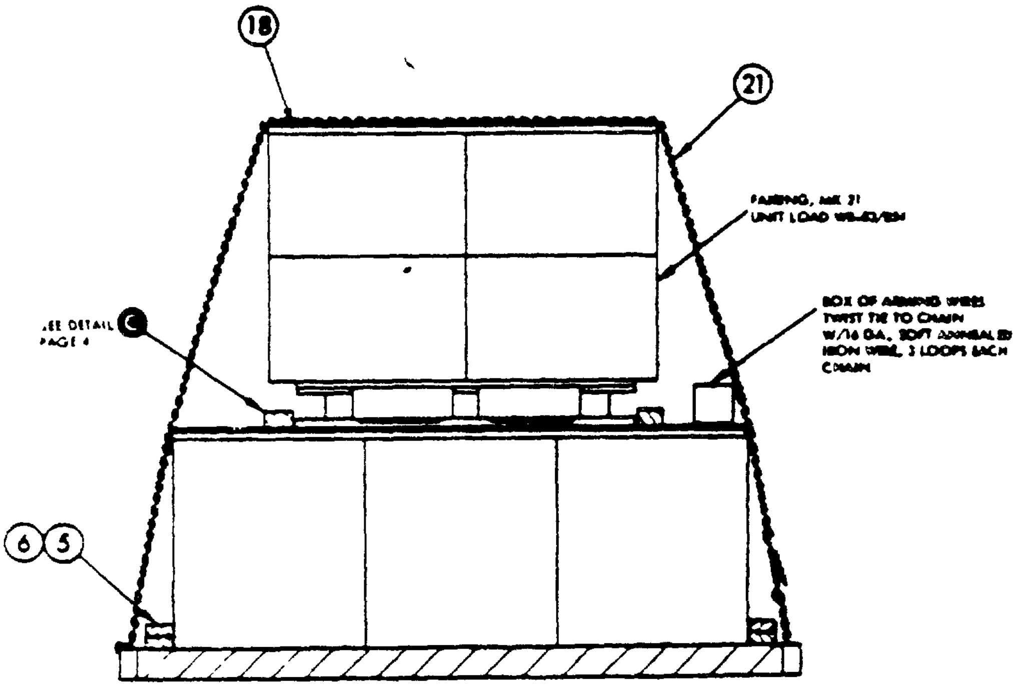


**DETAIL D**

PALLET BASE
(FOR PARACHUTE RELEASE MECH [illegible]
[illegible] 30/700)
1 REQUIRED



SECTION E-E



SECTION F-F

REVIEW ACTIVITY

NAVY - OS

PREPARING ACTIVITY

NAVY - OS
PROJECT NO 8140-N